Comparative Text Simplification using BART-large and T5 models on ASSET and TurkCorpus

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1. Introduction

This project focuses on evaluating the effectiveness of fine-tuning large language models for sentence simplification, using benchmark datasets and comparing the results against state-of-the-art (SOTA) systems reported in recent literature. The objective is to determine whether our fine-tuned versions of models such as T5 and BART can match or exceed existing SOTA performance in terms of simplification quality.

2. Fine-tuning Pretrained Models on ASSET and TurkCorpus

Fine-tuning refers to the process of continuing the training of a general-purpose language model on a task-specific dataset, in our case, simplification of complex english sentences.
We fine-tuned the pretrained models T5-small, T5-base and BART-large using the ASSET and TurkCorpus datasets.
We implemented the fine-tuning using the Hugging Face transformers library and the Seq2SeqTrainer API.

3. Feature Engineering and Deep Learning

In contrast to traditional machine learning models, deep learning approaches like T5 and BART do not require manual feature engineering. These transformer-based architectures automatically learn meaningful representations from the data through pretraining and fine-tuning. Therefore, we did not manually design features or apply dimensionality reduction techniques.
Our use of transformer-based models represents an end-to-end learning approach where the entire simplification pipeline (from encoding complex inputs to generating simplified output) is handled by a single model. Compared to earlier rule-based or hybrid systems, these deep learners perform significantly better with fewer assumptions and less task-specific tuning.

4. Dataset Insights

Based on our experiments, we propose the following recommendations for novices when analyzing these datasets:

- ASSET contains multiple reference simplifications per source sentence, which makes it more suitable for evaluation with metrics like SARI. TurkCorpus, in contrast, typically has fewer references and may reflect more stylistic variability.
- Both datasets are relatively small (~2,000 examples), which means that overfitting can be an issue. Fine-tuning large models should be done carefully with regularization or early stopping.
- Start with strong pretrained models like T5-base or BART to benefit from general linguistic knowledge.
- Fine-tuning using full train or test splits as provided, even small data can yield strong improvements.
- Compare with literature and leaderboard benchmarks, such as those listed on Papers With Code, to track progress.

5. Evaluation Criteria

5.1. Model Effectiveness

We evaluated the model's performance using three widely accepted metrics in text simplification:

- SARI: The primary metric, as it best correlates with human evaluation, accounting for Additions, Deletions, and Keeps.
- BLEU: Included as a secondary metric, though known to over-penalize simplifications that diverge lexically from references.
- FKGL (Flesch-Kincaid Grade Level): A readability measure; lower values indicate simpler output.
- We did not conduct full cross-validation due to resource constraints. Instead, we fine-tuned and directly evaluated the ASSET and TurkCorpus dataset.

5.2. Model Complexity

In this project, we employed three state-of-the-art transformer-based sequence-to-sequence models for the task of text simplification: BART-large, T5-base, and T5-small.

BART (Bidirectional and Auto-Regressive Transformers) is a denoising sequence-to-sequence model introduced by Lewis et al. (2020). Its architecture combines a bidirectional encoder, similar to BERT, with an autoregressive decoder, similar to GPT. BART is pre-trained by corrupting text with noise and learning to reconstruct the original text. This pre-training makes BART particularly suitable for generation tasks that require paraphrasing, summarization, and text simplification. The BART-large model contains approximately 406 million parameters, distributed across 12 encoder layers and 12 decoder layers, with hidden size 1024 and 16 attention heads. Due to its large size and denoising pre-training objective, BART-large is highly capable of capturing complex syntactic and semantic transformations, making it effective for simplification tasks.

T5 (Text-To-Text Transfer Transformer) was proposed by Raffel et al. (2020) as a unified framework for all NLP tasks formulated as text-to-text problems. In T5, both inputs and outputs are treated as text strings, allowing tasks such as translation, summarization, classification, and simplification to share a common architecture and objective. T5 models are pre-trained on a large-scale multi-task mixture called C4, using a span corruption objective. The T5-base model used in our experiments has approximately 220 million parameters, consisting of 12 encoder layers and 12 decoder layers, with hidden size 768 and 12 attention heads. Meanwhile, T5-small is a reduced version with approximately 60 million parameters, containing 6 encoder layers and 6 decoder layers, hidden size 512, and 8 attention heads. While T5-base offers a balance between performance and computational efficiency, T5-small is more suitable for low-resource or real-time scenarios where smaller models are required.

5.3. Comparison to state-of-the-art

Our outputs were as follows:

Model	Dataset	SARI	BLEU	FKGL
T5-base	TurkCorpus	30.974	97.004	9.533
	ASSET	37.138	92.971	8.424
T5-small	TurkCorpus	31.395	95.065	9.633
	ASSET	34.030	89.992	7.850
BART-large	TurkCorpus	32.149	96.030	9.352
	ASSET	42.304	97.072	8.362

As of the most recent Papers with Code leaderboard for sentence simplification on the ASSET dataset, the top performing models are:

Model	SARI	BLEU	FKGL
GPT-175B	47.94	73.92	7.73
MUSS (BART+ASCCESS Supervised)	44.15	72.98	6.05
Control Prefixes (BART)	43.58	(not given)	5.97

The leaderboard for sentence simplification on the TurkCorpus shows the following models:

Model	SARI	BLEU	FKGL
GPT-175B	47.46	73.83	9.33
MUSS (BART+ASCCESS Supervised)	42.53	78.17	7.60
Control Prefixes (BART)	42.32	(not given)	7.74

In comparison, our fine-tuned BART-large model achieved a SARI score of 42.3, BLEU of 97.07 and FKGL of 8.36, which is competitive with these recent and often more complex models. Unlike many leaderboards on multi-models or control mechanisms, our approach is based on a straightforward fine-tuning of publicly available transformer models.

When comparing our T5-base model to current state-of-the-art approaches like GPT-175B and MUSS, we observed that our model performed very well in terms of BLEU, even surpassing those more complex systems. This means our output was highly fluent and closely aligned with the human references.

Although our SARI and FKGL scores were slightly lower, as we expected, we used a much simpler setup– just a single pretrained model with basic fine-tuning, no extra components like control tokens or ensemble methods.

Despite using smaller and simpler models, our approach comes very close to SOTA in simplification quality, particularly in BLEU and fluency. With additional tuning or hybrid techniques, our models could likely reach or exceed leaderboard performance while remaining accessible and efficient.

5.4. Comparison with Peer Implementation

We also compared our results to Team number 2, working on the same task, but using the TurkCorpus dataset only.

Model	SARI	BLEU	FKGL
T5-Small (Team 8)	29.244	96.286	9.618
T5-Small (ours)	31.395	95.065	9.633

When comparing our T5-small model to Team 8's, we achieved a higher SARI score (31.4 vs. 29.2), which suggests our model made more effective and meaningful simplifications. Team 8 had a higher BLEU, which typically indicates closer word overlap with the references, but this can sometimes mean the model made fewer simplifications.

Their FKGL was slightly lower, meaning their output was a bit easier to read, though the difference is minimal. Overall, our model focused more on making real simplifications, which is better reflected in the SARI metric, the most important for this task.

5.5. Creative ideas

For both datasets, we applied consistent light preprocessing:

- Lowercased all sentences.
- Trimmed extra whitespaces.
- Filtered out training examples where the source sentence exceeded 100 words.

This preprocessing helped reduce noise and ensured consistency across datasets.

5.6. Domain Knowledge

In this project, we applied domain-specific knowledge about text simplification to guide our modeling and evaluation decisions. First, we prioritized the use of the SARI metric over BLEU, as SARI better reflects the quality of simplification based on human judgment. We also filtered out overly long source sentences (more than 100 words), based on prior understanding that models tend to perform poorly on very long inputs. Additionally, we avoided manual feature engineering, consistent with modern best practices for using end-to-end deep learning models for sequence generation tasks.

6. Conclusion

In this project, we demonstrated the effectiveness of fine-tuning publicly available transformer models—T5-small, T5-base, and BART-large—on benchmark datasets for sentence simplification, namely ASSET and TurkCorpus. Through empirical evaluation using SARI, BLEU, and FKGL, we showed that even with minimal preprocessing and relatively small datasets, transformer-based models can deliver highly competitive results.

Among the tested models, BART-large fine-tuned on ASSET achieved the highest overall performance, reaching a SARI score of 42.3, which closely approaches the best results reported on leaderboards. T5-base and T5-small also delivered strong results, especially in BLEU and readability. Notably, T5-small on TurkCorpus still produced high-quality outputs despite being significantly lighter, making it a viable option for low-resource settings.

7. References

ASSET Benchmark (Text Simplification). (n.d.). Papers with Code.
https://paperswithcode.com/sota/text-simplification-on-asset

Papers with Code. (n.d.). *TurkCorpus Benchmark (Text Simplification)*. Papers with Code.
https://paperswithcode.com/sota/text-simplification-on-turkcorpus

Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., ... & Zettlemoyer, L. (2020).
BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension.
In *Proceedings of ACL 2020*.
DOI: https://doi.org/10.18653/v1/2020.acl-main.703

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... & Liu, P. J. (2020).
Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer.
In *Journal of Machine Learning Research (JMLR), 21(140)*.
https://jmlr.org/papers/v21/20-074.html

Trainer — transformers 4.2.0 documentation. (n.d.). Hugging Face.
https://huggingface.co/transformers/v4.2.2/main_classes/trainer.html